ALPHAS: Adaptive Bitrate Ladder Optimization for Multi-Live Video Streaming

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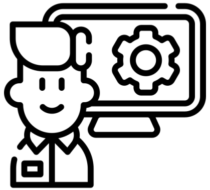
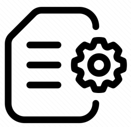
ATHENA *Adaptive Streaming over HTTP and Emerging Networked Multimedia Services*

IEEE INFOCOM 19–22 May 2025 London, United Kingdom

Live Video Streaming Pipeline



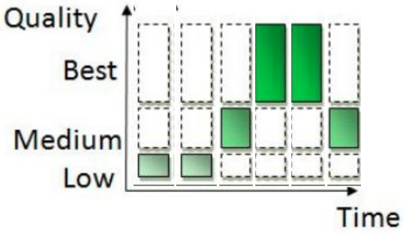
System Admin

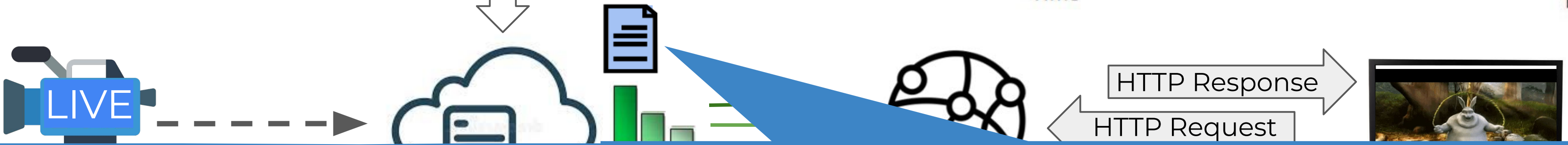


Bitrate Ladder

Bitrate	Resolution
5 Mbps	1920x1080
2.8 Mbps	1280x720
1.1 Mbps	960x540

Time



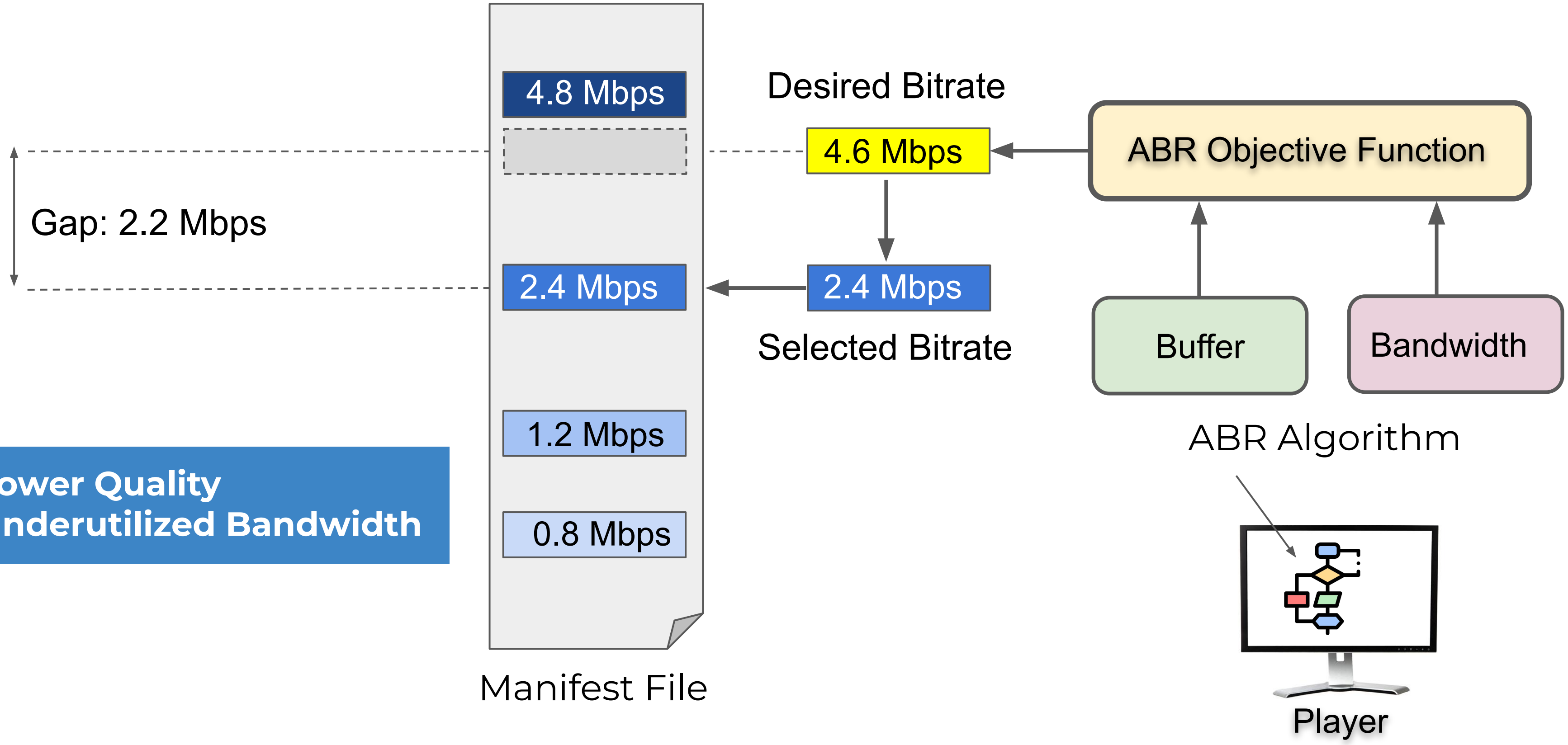


Li

- Fixed bitrate ladder
  - Content-aware, e.g., Netfli
  - Context-aware, e.g., [Lebre
  - Agnostic of the content aı

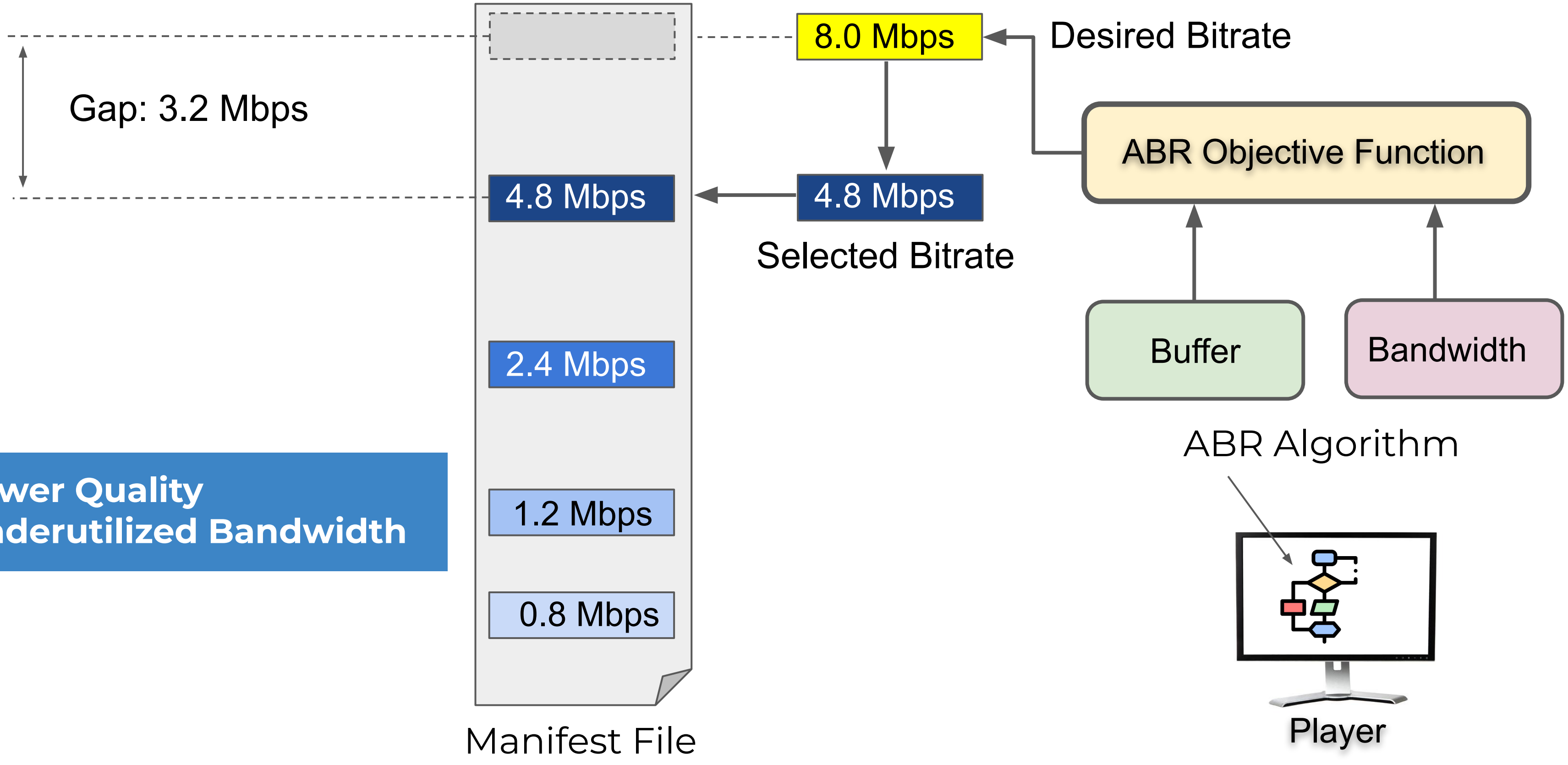
Manifest includes representations: different versions of the content optimized for various playback conditions, e.g., **different bitrates** and **resolutions**

Bitrate Selection by an Adaptive Bitrate (ABR) Algorithm



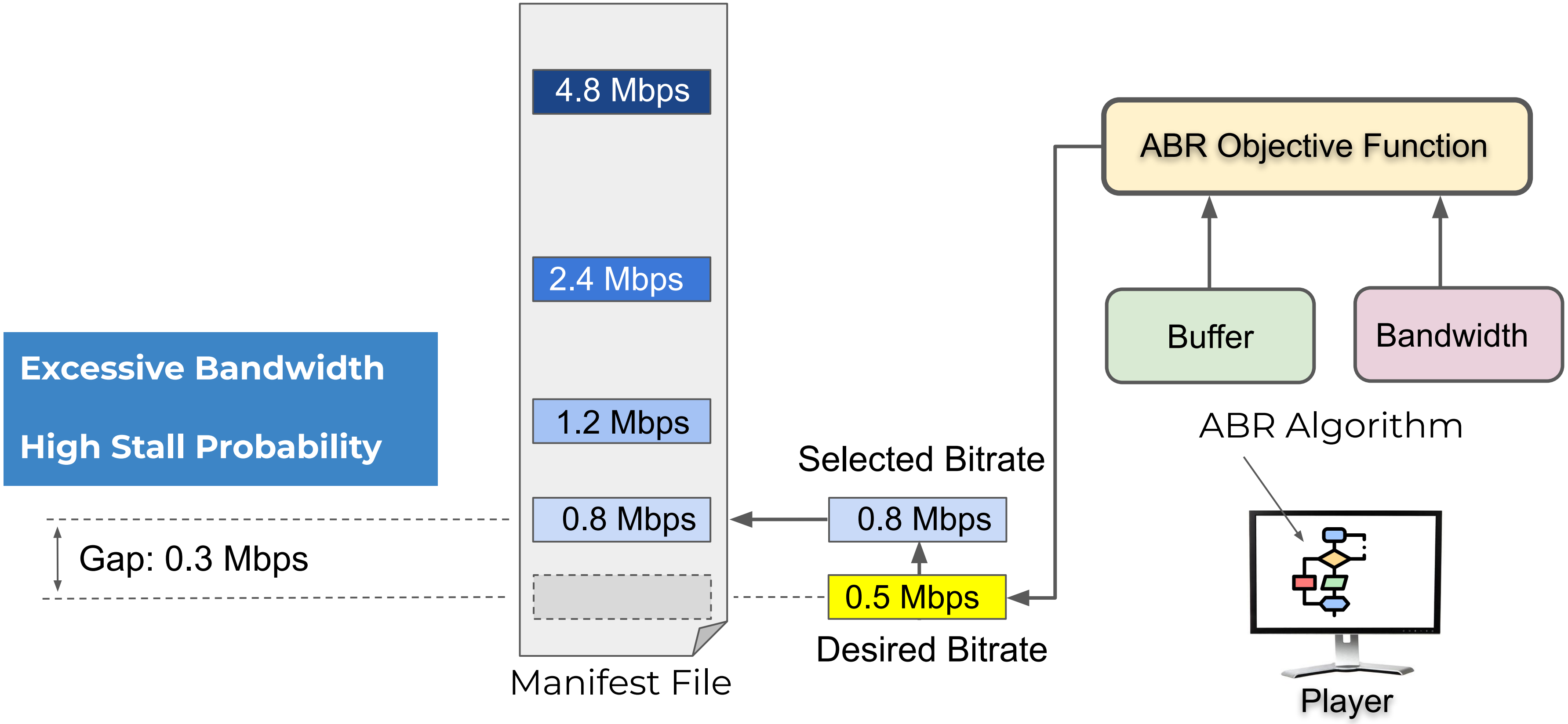
Lower Quality
Underutilized Bandwidth

The Highest Bitrate Might be Too Low



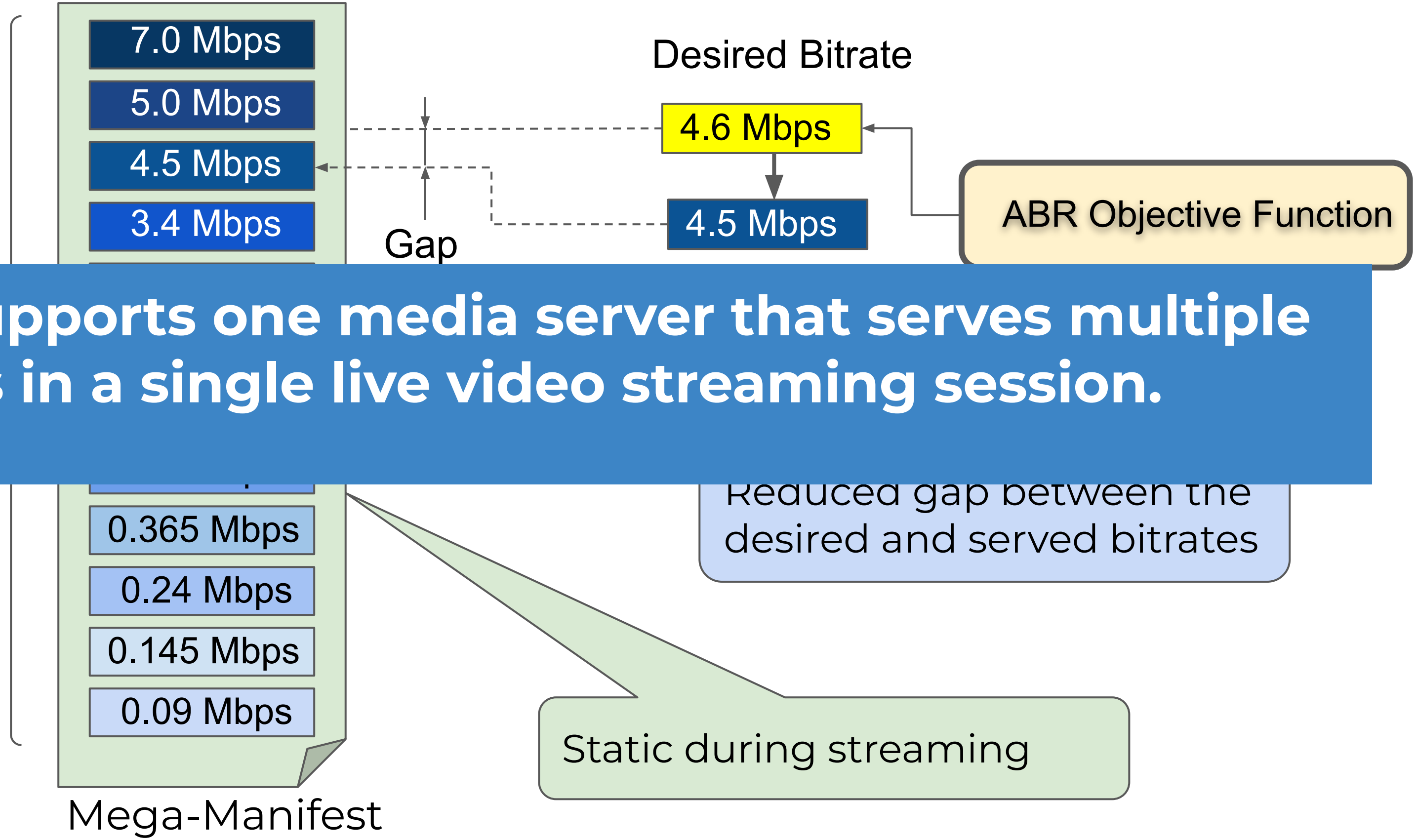
Lower Quality
Underutilized Bandwidth

The Lowest Bitrate Might be Too High



Mega-Manifest for Better Bitrate Alignment

ARTEMIS (NSDI 2024)*



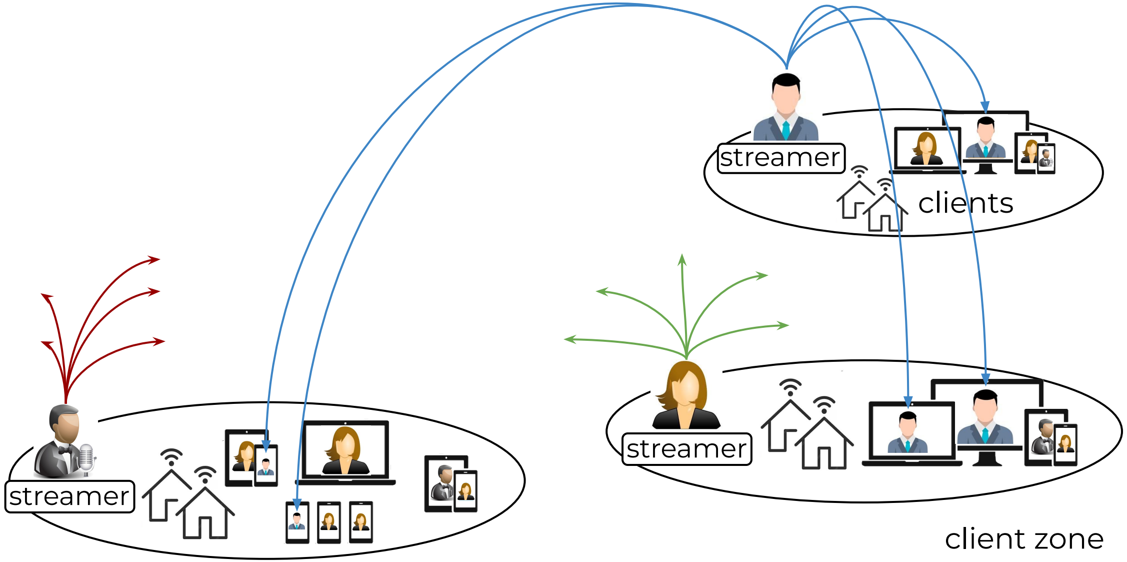
ARTEMIS supports one media server that serves multiple clients in a single live video streaming session.

ARTEMIS

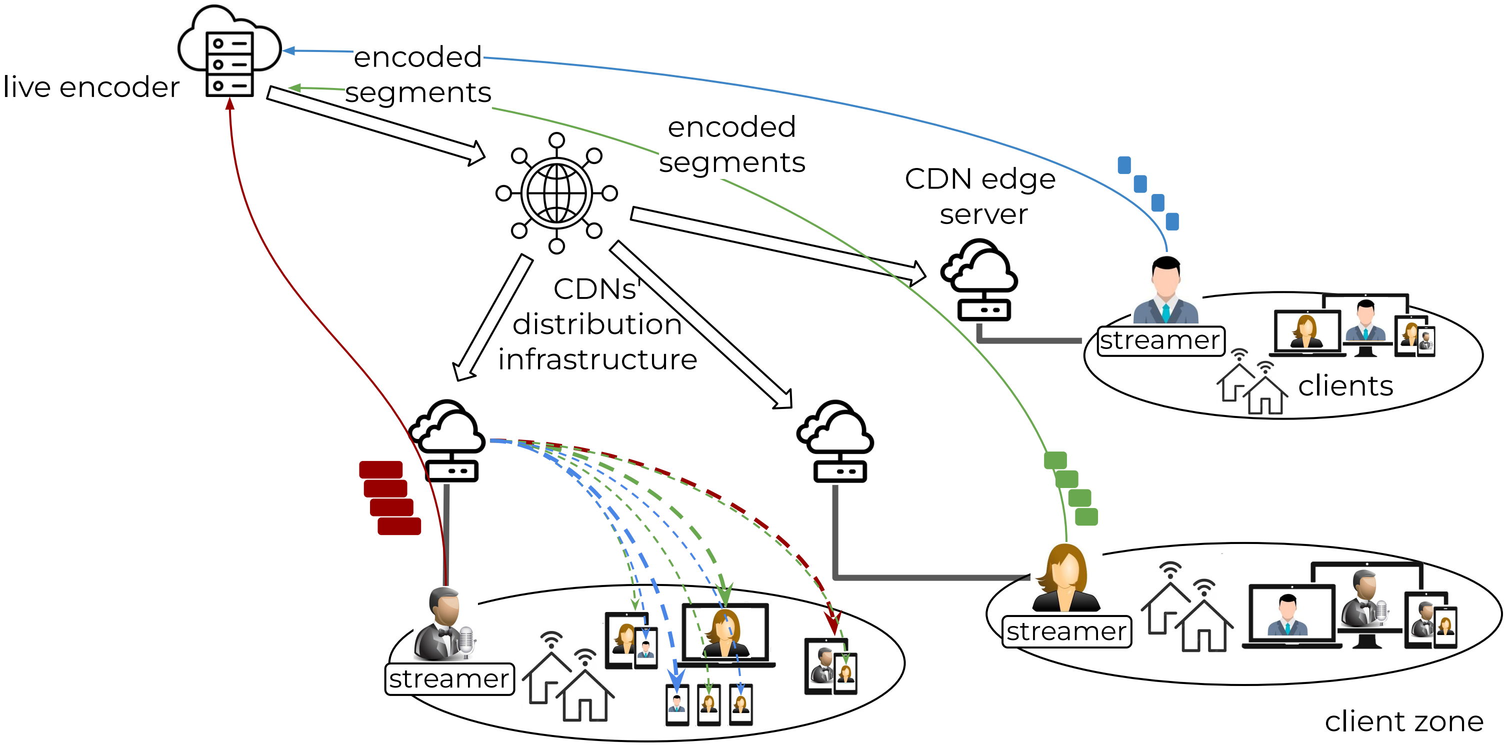
*** Tashtarian et al., "ARTEMIS: Adaptive Bitrate Ladder Optimization for Live Video Streaming", ACM NSDI 2024.**

Multi-live Video Streaming

Streaming live content from multiple streamers to heterogeneous subscribers across different zones.

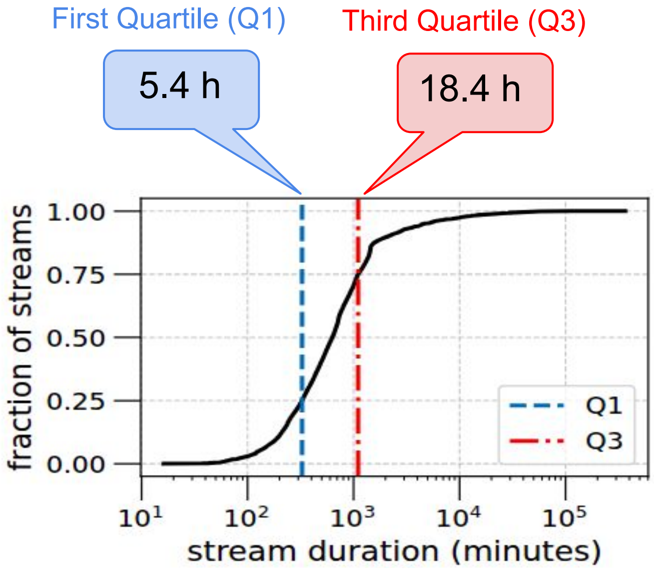


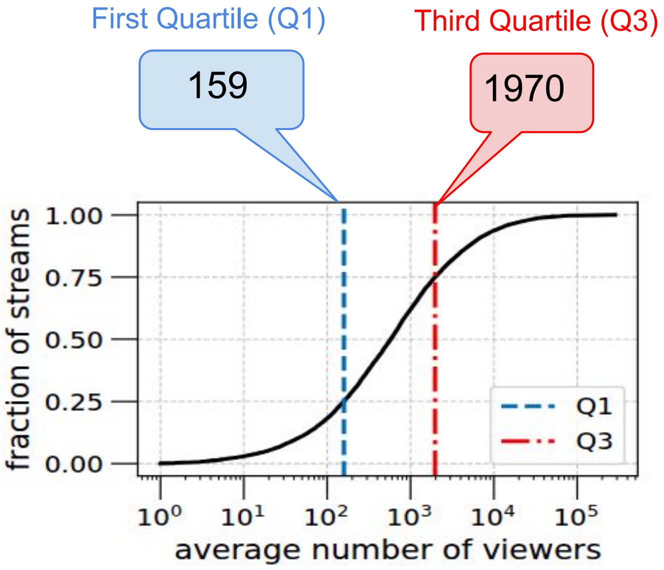
How can we determine appropriate bitrate ladder(s) for multi-live video streaming?



ALPHAS Motivation

Many live streams are long and exhibit significant variation in duration and popularity.

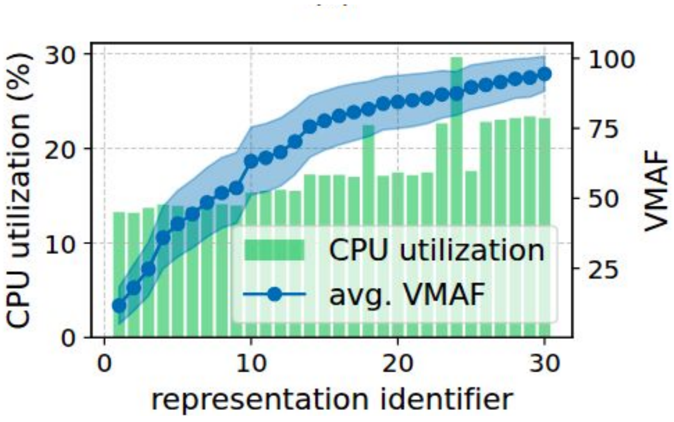


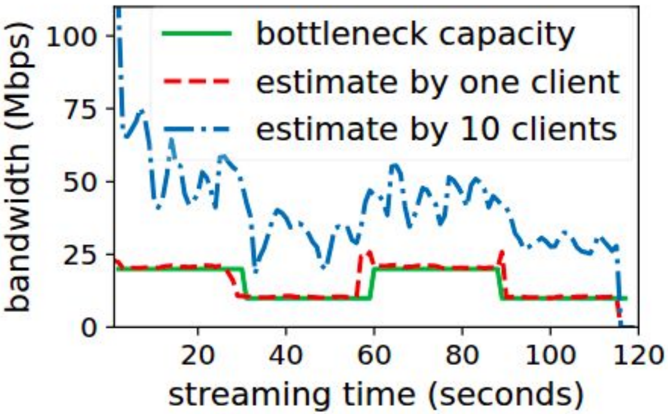


Based on collected real data on the stream duration and average number of concurrent viewers for over **12,000 live streams** on **YouTube** during **May and June 2024**.

ALPHAS Motivation …

Multi-live streaming needs to coordinate bandwidth usage, manage encoder resources, and prioritize streams

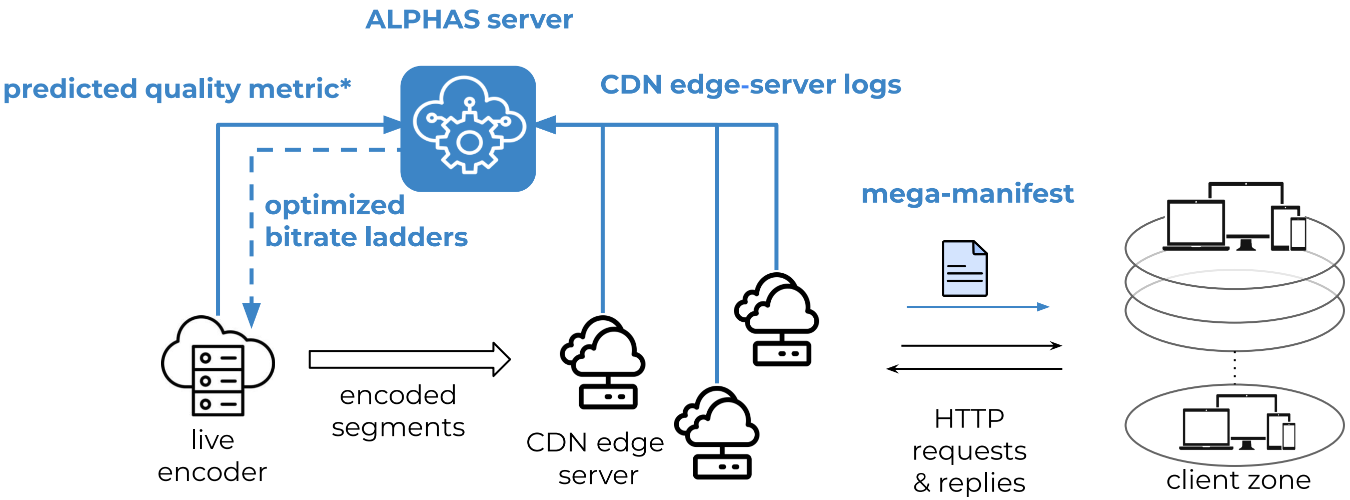




- Big Buck Bunny Content (30 representations)
- AWS EC2 t2.2xlarge instance
- Ultra-fast preset | FFmpeg

- Bandwidth between two AWS EC2 changes between 10 Mbps and 20 Mbps
- Two scenarios (1 and 10 clients)

ALPHAS Architecture



* H. Amirpour, J. Zhu, P. Le Callet, and C. Timmerer, “A Real-Time Video Quality Metric for HTTP Adaptive Streaming,” ICASSP, 2024.

ALPHAS Optimization Model: Integer Linear Programming (ILP)

1 $$\sum_{p \in \Omega, p \leqslant q} y^{v}_{p,q} = 1 \qquad \forall v \in V, q \in \Omega$$

ALPHAS serves each client request in the requested or lower rep.

Use of representation p to serve the clients of stream v that request representation q

ALPHAS Optimization Model: Integer Linear Programming (ILP)

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2 $\sum_{q \in \Omega, q \geqslant p} y^{v}_{p,q} \leqslant x_{v,p} \times |\Omega| \quad \forall v \in V, p \in \Omega$

Encoder must generate seg. of stream v in rep. *p* if it is selected.

Presence of representation *p* in the bitrate ladder of stream *v*

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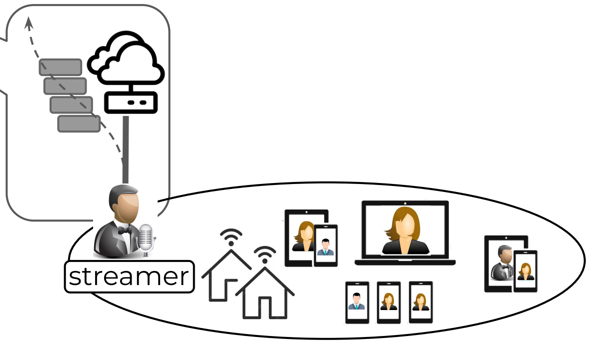
Encoder must generate seg of stream v in rep. p if it is selected.

3 $$\sum_{q > \hat{q}_v} x_{v,q} = 0 \quad \forall v \in V$$

Don’t generate bitrates exceeding the bitrate of the streamer supplied representation.



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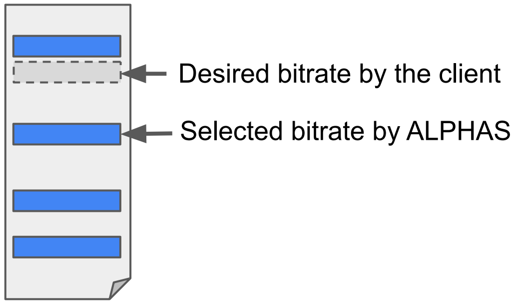
4 $\sum_{p < j \leqslant q} x_{v,j} \leqslant (1 - y^v_{p,q}) \times |\Omega| \ \forall v \in V, p, q \in \Omega, p < q$

Select the highest lower bitrate if the requested bitrate is not in the ladder.



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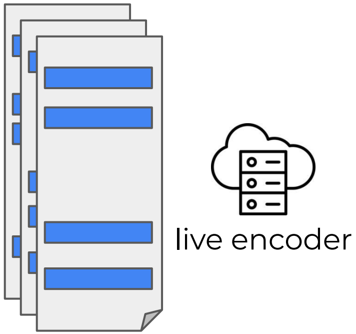
Select the highest lower bitrate if the requested bitrate is not in the ladder.

5 $$\sum_{v\in V} \sum_{q\in\Omega} \psi_q \cdot x_{v,q} \leqslant \Psi$$

Ensure the required computation capacity is available.



ALPHAS Optimization Model: Integer Linear Programming (ILP)

1. $\sum_{p\in\Omega,p\leqslant q} y^v_{p,q} = 1 \quad \forall v\in V, q\in\Omega$ — ALPHAS serves each client request in the requested or lower rep.

2. $\sum_{q\in\Omega,q\geqslant p} y^v_{p,q} \leqslant x_{v,p} \times |\Omega| \quad \forall v\in V, p\in\Omega$ — Encoder must generate seg of stream v in rep. p if it is selected.

3. $\sum_{q>\hat{q}_v} x_{v,q} = 0 \quad \forall v\in V$ — Don't generate bitrates exceeding the bitrate of the streamer supplied representation.

4. $\sum_{p<j\leqslant q} x_{v,j} \leqslant (1-y^v_{p,q}) \times |\Omega| \quad \forall v\in V, p,q\in\Omega, p<q$ — Select the highest lower bitrate if the requested bitrate is not in the ladder.

5. $\sum_{v\in V}\sum_{q\in\Omega} \psi_q \cdot x_{v,q} \leqslant \Psi$ — Ensure the required computation capacity is available.

6. $\sum_{v\in V}\sum_{q\in\Omega} r^{z,v}_q \left(\sum_{p\in\Omega,p\leqslant q} \rho_p \cdot y^v_{p,q}\right) \leqslant D_z \quad \forall z\in Z$ — Ensure the CDN edge server's bandwidth is available.

ALPHAS Optimization Model: Integer Linear Programming (ILP)

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Ensure the CDN edge server’s bandwidth is available.

7 $$\sum_{q\in\Omega} x_{v,q} \leqslant \ell \quad \forall v\in V$$

Limit the number of representations in a bitrate ladder

ALPHAS Optimization Model: Integer Linear Programming (ILP)

OPT-ALPHAS

$$\max_{\boldsymbol{x},\boldsymbol{y}}. \ f(\boldsymbol{y})$$

$$f(\boldsymbol{y}) = \sum_{z \in Z} \sum_{v \in V} p_{z,v} \cdot \overline{y}_{z,v}$$

Priority of stream v in zone z

$$\overline{y}_{z,v} \leqslant \frac{1}{C_{z,v}} \sum_{q \in \Omega} r_q^{z,v} \left(\sum_{p \in \Omega, p \leqslant q} \nu_{v,p} \cdot y_{p,q}^{v} \right)$$

The average video quality for the clients of stream v in zone z.

Subject to

Constraint 1 through 7

Binary variables

- $\boldsymbol{x}$ Representation q in the bitrate ladder of stream v
- $\boldsymbol{y}$ Representation p to serve the clients of stream v that request representation q

Heuristic Algorithm: Submodular ALPHAS (SM-ALPHAS),

SM-ALPHAS

$$\mathcal{I} = \{(v, q) \mid v \in V, q \in \Omega\}$$

pair stream v and representation q

$$\tilde{f}(\mathcal{J}) = \max_{\boldsymbol{y}}.\ f(\boldsymbol{y})$$

Subject to: Constraints 1 through 4

$$x_{v,q} = 1 \leftrightarrow (v, q) \in \mathcal{J} \subseteq \mathcal{I}$$

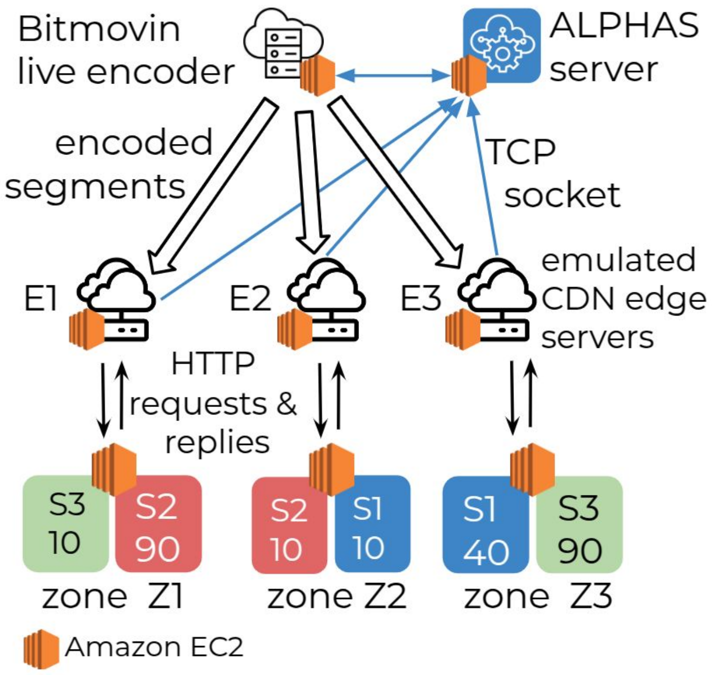
Theorem 1. $\tilde{f}(\mathcal{J})$ is a submodular set function.

Theorem 2. SM-ALPHAS computes a feasible solution to the multi-live streaming problem.

Theorem 3. SM-ALPHAS executes in polynomial time.

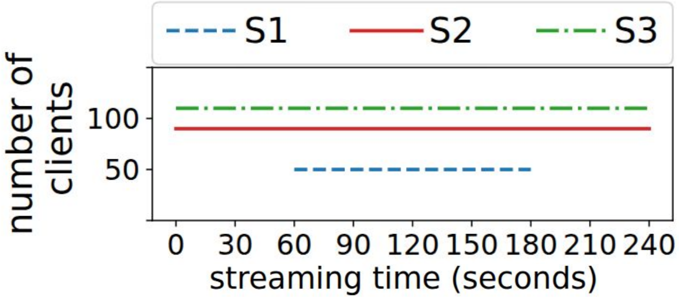
⋮

Experimental Setup

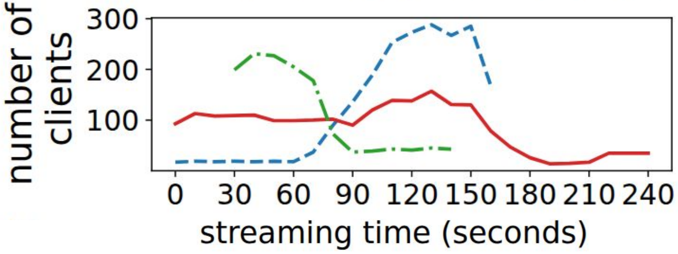


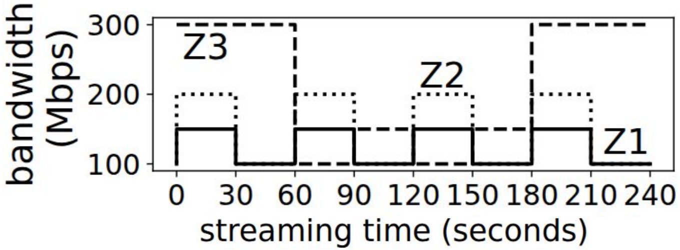
Cloud-based implementation of ALPHAS

Static Subscription



Dynamic Subscription





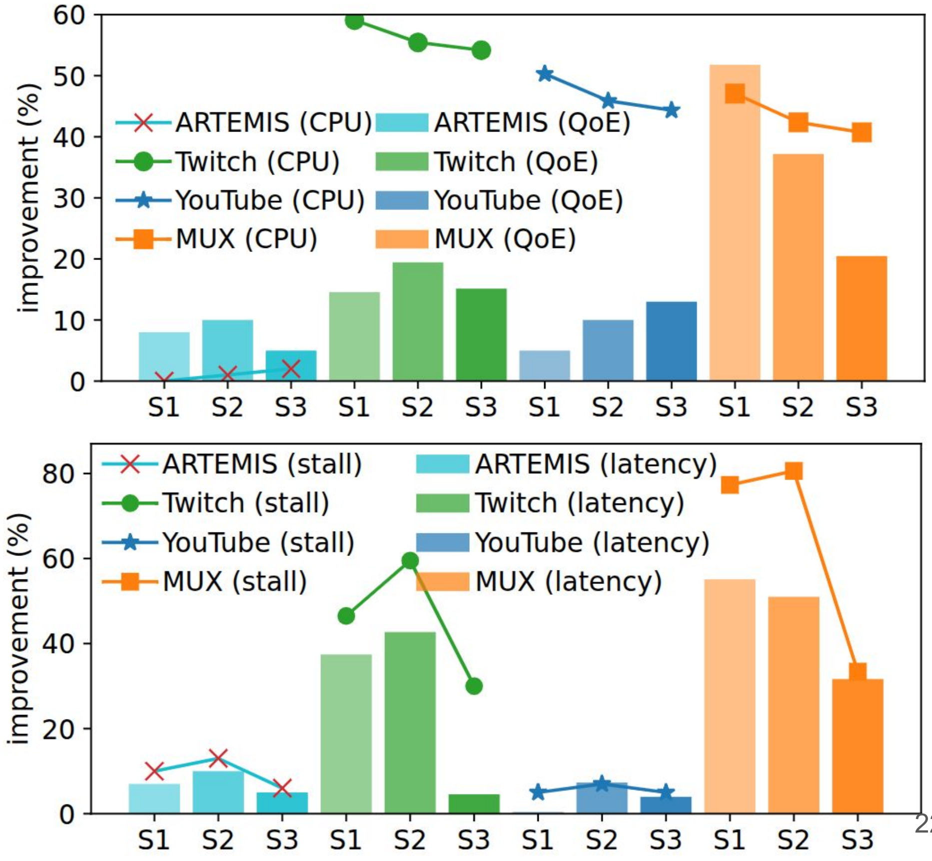
Improvement by OPT-ALPHAS

CPU utilization and QoE

OPT-ALPHAS improves QoE by up to **55%** for MUX and reduces CPU utilization by over **50%** for Twitch, demonstrating its dual benefit on performance and user experience.

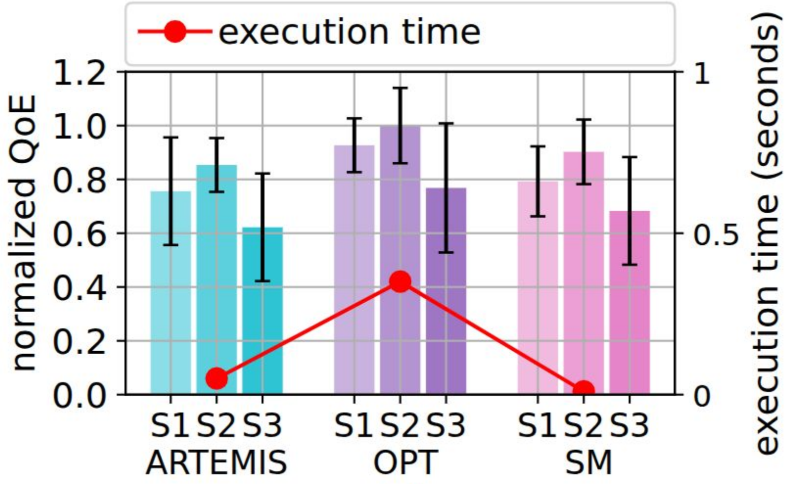
Stall duration and end-to-end latency

OPT-ALPHAS reduces stall duration by up to **70%** for MUX and **60%** for Twitch, while cutting end-to-end latency by around **40–75%**, significantly enhancing real-time streaming quality.



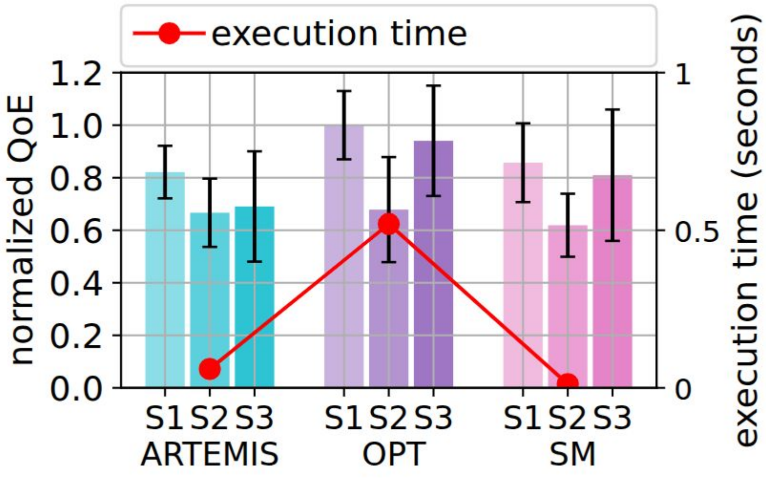
Average Normalized QoE and Execution time

OPT-ALPHAS achieves the highest QoE across all scenarios with a modest execution time (~0.4s), outperforming ARTEMIS and SM-ALPHAS in both quality and efficiency.



Identical Priorities

Even under different priority settings, OPT-ALPHAS maintains high QoE while keeping execution time below 0.6s. QoE values follow the assign priorities in APLHAS.



Different Priorities
(S1 and S3 have higher priority than S2)

Conclusion

- **ALPHAS** addresses the critical challenge of bitrate ladder optimization in **multi-live video streaming**.
- Our **ILP-based OPT-ALPHAS** approach significantly enhances:
  - **QoE** (up to **55%** improvement for MUX)
  - **CPU utilization** (over **50%** reduction for Twitch)
  - **Latency** (up to **75%** reduction)
  - **Stall duration** (up to **70%** reduction)

- Our **heuristic SM-ALPHAS** guarantees polynomial-time feasibility with close-to-optimal performance.
- ALPHAS enables **scalable, efficient, and high-quality multi-stream live video delivery**.

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